Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/02/2011
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Project Title:	Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2012 per NSSC information. Change in Risk/Gaps per HRR information (Ed., 9/26/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Clinton (SUNY Stony Brook)		
Grant/Contract No.:	NNX08BA35G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	As space missions become longer, it is apparent that long-term risks of multiple, or even single, exposures to microgravity on crew musculoskeletal health need to be defined and addressed. Even though the magnitude of the long-term deterioration in morphology and strength is largely unknown, it is clear that such risks will be heavily influenced by the genetic make-up of an individual. We have recently identified and characterized a genetically diverse mouse population that, similar to astronauts, includes a diverse range of responses to hindlimb unloading and reambulation. Using this population, here, we will address the consequences of multiple exposures to simulated weightlessness on long-term quantity and quality of trabecular and cortical bone as well as muscle, and investigate how recovery periods between periods of weightlessness can be optimized. Changes in tissue morphology, density, and mechanical properties will be assessed longitudinally in each genetically distinct mouse by in vivo microCT and		

	finite-element modeling, allowing for the identification of skeletal baseline indices that modulate changes in tissue morphology and strength during altered levels of loading. Detailed mechanical tests will be performed at the endpoints to assess the long-term quality of the tissue. Together, the proposed studies provide a unique opportunity to establish the range of responses of genetically diverse individuals to multiple unloading and reambulation phases. Optimizing the length and distribution of recovery periods between disuse exposures may ultimately help to maintain long-term musculoskeletal morphology and strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Extended periods of unloading are not confined to space flight but also occur during extended bedrest or paralysis on Earth. Therefore, data from our experiments may help understanding the basis by which the musculoskeleton responds to repeated unloading events and eventually lead to more efficacious prophylactics and treatments.
Task Progress:	Introduction The magnitude and rate of bone atrophy that accompanies the loss of functional weightbearing during space flight is staggering -- on average, astronauts lose bone mineral in the lower appendicular skeleton at a rate of approximately 1-3% per month. The atrophy is site-specific and type-specific, with trabecular bone loss 3-5 times greater than cortical bone loss. Since the extent of bone atrophy is modulated by age, gender, genetics, and the duration of unloading, a comprehensive risk assessment requires consideration of all these factors. Though a single exposure to microgravity is known to result in significant bone loss, the musculoskeletal consequences to multiple exposures are largely unknown. Since disuse alters bone quantity, quality, and remodeling, factors known to influence bone's response to catabolic signals, we hypothesize that prior exposures to disuse should affect the response to subsequent exposures. Using a genetically heterogeneous F2 mouse population (BALB/cByJ x C3H/HeJ) which exhibits a highly variable response to hindlimb unloading (HLU) and reambulation (RA), we are investigating the consequences of multiple exposures to simulated microgravity on long-term bone quantity, quality, and the strength of trabecular and cortical bone, and studying how the length and distribution of recovery periods between periods of weightlessness can be optimized to minimize musculoskeletal risks. Hypothesis 1: Multiple short exposures to disuse are less detrimental to musculoskeletal morphology and mechanical strength than a single long exposure. Specific Aim 1 (SA1): To determine whether the total exposure time to weightlessness conditions or the number of repetitive exposures is the primary determinant of skeletal morphology and strength. Hypothesis 2: Multiple exposures to disuse cause long-term defects in musculoskeletal morphology and mechanical properties, but longer recovery periods in between multiple microgravity exposures can mitigate this decline. Specific Aim 2 (SA2): To determine the musculoskeletal consequences of altering the length and distribution of the recovery periods between multiple exposures to weightlessness. Progress Year 3 We have completed the first specific aim and a pilot study in which we studied the influence of age on bone's response to multiple exposures. Building on data described in the previous two progress reports, three manuscripts are currently either in review or in preparation. The second specific aim is ongoing and progress is described in greater detail below. Animals Ninety adult (16wk) female F2 (BALB/cByJ x C3H/HeJ) mice were divided into three groups (n=30/group) – two groups subject to disuse and an age-matched ambulatory control. The experimental disuse groups were exposed to a total of 9wk of HLU and 42wk of RA. In contrast to SA1, all disuse groups were subjected to three 3wk HLU cycles. However, the RA period following each HLU period was either 3wk or 9wk. The physical activity of each mouse was monitored using a 3-dimensional activity monitoring cages for 24 hours at baseline, before and after each HLU cycle, and before sacrifice. Animals were single-housed and given free access to food and water during ambulation, HLU, and RA. Animals were weighed and randomly distributed into the different experimental groups. The well-characterized Morey-Holten hindlimb unloading model was used to simulate microgravity in all studies. All animals were sacrificed at 14mo of age, about 6 months after the completion of the final HLU cycle. All mice have completed the HLU periods and are currently in the final RA period prior to sacrifice. Results Activity Patterns: Since the total activity count over a 24-hour period across all groups at baseline demonstrated that individual activity components maintained their proportions, we only present the sum of all activities below. Throughout the experimental duration, total activity counts were almost identical between the two experimental groups and not significantly from the control group. Abdominal Fat: Consistent with changes observed in astronauts, the temporal changes in abdominal fat consistently exhibited a reduction during HLU and an increase during RA. The 9wk recovery period was sufficient to normalize the loss in adiposity experienced during the first unloading period. In contrast, the 3wk recovery period was too short for fat levels to return to control levels. In this group, the decrease in fat volume during the second and third HLU period were much smaller than during the initial unloading phase, giving rise to almost identical total fat levels as in the 9wk recovery group at week 31 of age. Here, only total adipose tissue (TAT) volume is reported but the temporal patterns were indistinguishable from the changes in subcutaneous adipose tissue (SAT) and visceral adipose tissue (VAT) volume. Spinal Bone Density: The in vivo µCT scans of the abdomen facilitated the analysis of apparent volumetric density of the lumbar spine. In normally ambulating controls, bone density slightly increased until 40wk of age. In contrast, 3wk of HLU greatly reduced bone density. Similar to changes in adipose tissue, the 9wk but not the 3wk reambulation period restored bone density to control levels. In the 3wk recovery group, the magnitude of bone's response to unloading lessened with the second and third unloading period, rendering a spinal bone density by 31wk of age that was similar to

that of the 9wk recovery group in which the effect of the second unloading period was similar to the first one.

Distal Metaphysis of the Femur: Trabecular bone volume fraction (BV/TV) in normally ambulating control mice was relatively stable over the experimental period. As expected, both experimental groups experienced a large drop in bone volume fraction over the first unloading period. They also both started to recover some of these losses during the initial reambulation phase. The degree of recovery was almost 3x as great in the 9wk recovery group than in the 3wk recovery group. Three interesting phenomena were observed. Firstly, the drop in bone volume fraction was greatly reduced during the second and third unloading phase in the 3wk recovery group but not in the 9wk recovery group. Secondly, after the second unloading phase, the 3wk group did not recover any bone during the subsequent reambulation phase. And thirdly, the total amount of bone loss was almost identical between the two experimental groups by wk31 and significantly different from normal controls. Bone's susceptibility to unloading and reambulation was not only reflected by changes in the quantity of trabecular bone, but also by its micro-architecture. As revealed by changes in normally ambulating mice, aging caused a steady decrease in the number of trabeculae which concomitantly became thicker. Unloading caused significant decreases in these two architectural parameters. During reambulation, trabecular thickness increased while trabecular number continued to decrease. Similar to all other outcome measures presented above, the values for trabecular number were similar between the 3wk and 9wk recovery groups at the end of the second (third) HLU period (wk 31 of age). Trabecular thickness was the only variable that was different between the 3wk and 9wk recovery groups. Interestingly, including a 3wk recovery period after unloading was more beneficial than the 9wk recovery period. Trabecular tissue mineral density, while sensitive to HLU and RA, only showed very small changes during the loss and regain or weightbearing.

Cortical bone was also sensitive to unloading, however to a much lesser degree. Even though the magnitude of the response of cortical bone was less, the patterns were remarkably similar to trabecular bone; at 31wk of age, the total amount of cortical bone loss was almost identical between the two experimental groups.

Mechanical Modeling of the Femoral Distal Metaphysis: In vivo microcomputed tomography accurately tracks changes in trabcular and cortical bone quantity and micro-architecture but does not directly provide any data on bone's mechanical stiffness. To this end, we employed finite lement modeling (FEM) to convert the µCT scans into mechanical models. Qualitative preliminary results emphasize the devastating mechanical impact that alterations in trabecular and cortical bone morphology can have. In control bones, the relatively small, age-related decrease in bone volume did not significantly raise stress values between baseline and 31wk of age. In contrast, a significant increase in Von Mises stresses were observed in bones that underwent consecutive unloading/reambulation cycles, regardless of whether the recovery period was 3wk or 9wk. Quantitatively, as a function of age, peak von Mises stresses closely tracked the changes in trabecular bone volume fraction.

Correlations between Activity Patterns and Changes in Morphologic Indices: One of the motivations for expanding the scope of this grant proposal and to add activity monitoring was to address the question whether the differential capability of individual mice to recover tissue during reambulation is related to their level of activity. We have found so far that the amount of activity only plays a minor role in explaining the differences in regaining tissue after unloading (as reflected by R2 values that are all below 0.3).

Summary

The goal of this second specific aim is to determine whether extending the rest period between consecutive unloading cycles can attenuate the detrimental effects of the loss of weightbearing on the skeleton. The preliminary analysis presented here strongly suggests that a reambulation period that is three times as long as the actual unloading period is not beneficial compared to a reambulation period that is of equal length to the unloading period. As expected, outcome measures from all tissues analyzed so far indicate that a shorter reambulation period greatly limits the degree of tissue recovery. Interestingly, the data also consistently indicate that shortening the reambulation phase blunts the impact of the following unloading cycle. Together, these two trends cancelled each other out, effectively leading to similar tissue quantity and quality after multiple HLU/RA cycles irrespective of the actual duration of the recovery phase. The mice from the different groups are currently in their last reambulation phase that will determine the long-term impact of a 3-fold difference in recovery phase duration between consecutive loading cycles. Mice will be sacrificed in Dec 2011. It is fully expected that all bone/muscle/fat/activity level analyses will be completed by the end of the requested one-year no-cost extension period.

EDITOR'S NOTE (12/17/12): PROJECT CONTINUES AS "Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure--NNX12AH32G" WITH THE SAME PRINCIPAL INVESTIGATOR AND PERIOD OF PERFORMANCE OF 9/30/12-12/31/12. SEE THAT PROJECT FOR SUBSEQUENT REPORTING.

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